

--	--	--	--	--	--	--	--

***B.Tech. Degree VII Semester Special Supplementary Examination
May 2017***

**IT 1705 (E2) CRYPTOGRAPHY AND DATA SECURITY
(2012 Scheme)**

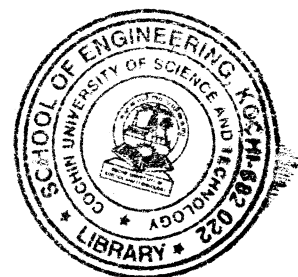
Time : 3 Hours

Maximum Marks : 100

PART A(Answer *ALL* questions)

(8 × 5 = 40)

- I. (a) Explain cryptography and cryptanalysis.
- (b) Define Coincidence Index (CI).
- (c) Explain characteristics of DES.
- (d) Briefly explain finite state machines.
- (e) Differentiate entity authentication and message authentication.
- (f) Explain knapsack problem.
- (g) Explain the general aspects of key management.
- (h) Briefly explain Diffie-Hellmann protocol for key exchange.

**PART B**

(4 × 15 = 60)

- II. (a) Explain various aspects of security. (8)
- (b) Explain transposition cipher system. (7)
- OR**
- III. (a) Explain the working of Hagelin machine with a neat diagram. (10)
- (b) Discuss various types of cryptanalytic attacks. (5)
- IV. Explain DES algorithm with needed diagrams. (15)
- OR**
- V. Explain International Data Encryption Algorithm (IDEA) in detail. (15)
- VI. (a) What do you mean by public key systems? Briefly explain RSA algorithm. (8)
- (b) Explain message integrity using hash functions. (7)
- OR**
- VII. Explain message authentication using digital signature. State digital signature algorithm. (15)
- VIII. How is the key distributed for asymmetrical systems? Explain. (15)
- OR**
- IX. (a) What do you mean by network security? What are the different ways of employing network security? (10)
- (b) Write short notes on fair crypto systems. (5)